

ROI with AI

The REAL question for Enterprises deploying AI

By Ashwin Rangan

Lately, there has been obsessive speculation and reporting on whether and when AI companies like OpenAI and Anthropic will achieve break-even, leave alone profitability. Answers have ranged from “never” to “within the next decade”. That range of answer has been underpinned by boundaries assumed by whoever has done the analysis. Frankly, this debate – while full of salacious content - should be of interest to only those people who are either employed by these companies or who wish to invest in these companies (both of which are still pre-public).

A related question in the press about ROI has been about the \$750 billion in capital spending, made in 2026 by the “hyperscalers” – Google, Meta, Amazon and Microsoft. I have made a compelling argument elsewhere that these are not at all speculative, so long as these are one-time capital expenditures. The hyperscalers will more than recover these outlays within the next 3 to 4 years, so long as their historic rates of growth hold-up. As with the example of OpenAI and Anthropic, this question should be of interest only to the employees of these hyperscalers and to investors who are anxiously eyeing press releases, detailing capital commitments.

The REAL question which almost all other businesses should worry about is:

Does YOUR company have an ROI with AI?

To be sure that we are addressing the right problem correctly, it is worth a short deviation to first understand the exact meaning of the term ROI. Investopedia defines ROI as a percentage measure of profitability that tells you how much gain or loss an investment generated relative to its cost. It answers a simple question: *“For every dollar I invested, how much did I earn (or lose)?”*

Investopedia goes on to say that ROI is the net profit divided by the cost of the investment, expressed as a percentage. It’s used across finance—capital projects, stocks, business ventures, and real estate—to evaluate efficiency and compare alternatives.

The exact formula, according to Britannica, is:

$$ROI = \frac{\text{Final Value} - \text{Initial Total Cost}}{\text{Initial Cost}} \times 100$$

This gives a percentage return.

For example: Invest \$5,000 and the returned final value is \$5,500.

$$ROI = \frac{5500 - 5000}{5000} \times 100$$

This gives a 10% ROI.

Step 1 in AI projects is getting a handle of Initial Total Cost.

With AI, there are several elements of Initial **Total** Cost which enterprises should recognize. These should include the cost of access to AI compute and storage infrastructure (appropriately apportioned to the project at hand); the talent which is deployed to build and operate the AI infrastructure for the purposes of the project; and the cost of tokens used in resolving whatever problem(s) are a part of the project.

The cost of Talent here should include not only direct Engineering labor, but also supporting labor – like Project Management, Executive talent etc. Of course, they should be pro-rated as required. In essence, these are “opportunity costs”.

These cost-elements add up and answer the question of Initial Cost. Getting a handle on this is just basic business.

Step 2 is the tricky part. It is getting a handle on the Final Value. This depends on a disciplined approach to computing this interesting number. That seems intuitively obvious, but there is a method to this approach.

The first lever is data-selection.

AI relies on data to “find better solutions”.

In every enterprise – whether for-profit or otherwise – there is a bank of data which is routinely gathered to conduct day-to-day business. In Walmart, it could be Inventory and merchandise data. In Dell Corporation, it could be specific configurations within customer segments. In Telecom companies, it could be customer data and billing data.

Selecting a juicy problem which is directly related to day-to-day data is a strategic way of approaching problem selection and AI deployment.

This is an inherent **second lever: strategic problem-selection.**

If this day-to-day core operational data is considered out-of-bounds for whatever reason, choosing datasets which are not as frequently accessed as day-to-day operational data could yield good results with AI treatment – only, the value is likely to be lower.

The third lever is problem-formulation and hypothesis testing.

Formulating the right problem has high leverage. For example, in Dell Corporation, using AI, analysis may reveal that certain configurations are more frequently ordered than most others.

Using those results, could Dell pre-configure those configurations, further accelerating time-to-delivery? And by accelerating it, could the Customer be convinced to pay a premium for earlier delivery? And if so, what is the incremental value of the premium? What percentage of orders could benefit?

These are testable attributes. Customers will self-select and provide answers to these questions, either through interview panels or with A/B testing.

Once the hypothesis has been tested and proven, **the fourth lever is value-testing.**

In this case, Final Value (FV) can be computed as:

$$FV = \text{Premium per unit} * \% \text{ of total units with those configurations}$$

If FV is greater than Initial Total Cost, the ROI is positive.

The final lever is hurdle-rate testing.

Most companies will have an internal hurdle-rate to assess ROI. Investopedia describes the hurdle rate as the minimum return required for an investment to be acceptable, adjusted for risk. If the ROI exceeds the hurdle rate, it is a clear winner!

When AI projects are approached with this sort of discipline, they will yield acceptable – if not startlingly high – ROI.

Without this discipline, AI will chase ROI.